Bit Manipulation

Bit → Binary Digit

$(11)_{10} \rightarrow 1011$ $(20)_{10}$

$(9)_{10} \leftarrow + \; 1001$

10100

$16 + 4 = 20$

Bitwise operation

① Binary AND or Bitwise AND &

② Binary OR |

③ Binary XOR ^

④ One's complement ~ (not operator)

⑤ Binary Left Shift <<

⑥ Binary Right shift >>

x	y	x & y	x | y	x ^ y
0	0	0	0	1
0	1	0	1	0
1	0	0	1	0
1	1	1	1	1

XOR

$x \oplus y = x\bar{y} + \bar{x}y$

x	~x
0	1
1	0

$10 \,\&\, 12 = $

$$\begin{array}{r} 1010 \\ \& \; 1100 \\ \hline (1000)_2 \end{array} = 8$$

$10 \,|\, 12 = 14$

$$\begin{array}{r} 1010 \\ \text{or} \; 1100 \\ \hline (1110)_2 \end{array} = 14.$$

$\sim(10110110)_2 = (01001001)_2 \xrightarrow[+1]{2^{nd}\text{ comp}} (01001010)_2$

$\xrightarrow{\text{1's comp}}$

$\leftarrow$

$(10111000)_2 \xrightarrow{2^{nd}\text{ comp}} (01001000)_2$

n bit

the range of +ve Integer = $[0, 2^n - 1]$

$\downarrow$ negative number

n = 3 $= [-3, 3]$

0 0 0	$\to +0$
0 0 1	$\to +1$
0 1 0	$\to +2$
0 1 1	$\to +3$
1 0 0	$\to -3$
1 0 1	$\to -2$
1 1 0	$\to -1$
1 1 1	$\to -0$

$\left.\begin{matrix} +0 = 000 \\ -0 = 111 \end{matrix}\right\} 0$

Range [negative + positive] $= [-2^{n-1}+1, \; 2^{n-1}-1]$

$\downarrow$

[1's complement]

Proper negative

$\downarrow$

2nd complement

3 bit integer

0 0 0	$\to 0$
0 0 1	$\to 1$
0 1 0	$\to 2$
0 1 1	$\to 3$
1 0 0	$\to -4$
1 0 1	$\to -3$
1 1 0	$\to -2$
1 1 1	$\to -1$

MSB = 0 $\to$ +ve

1 $\to$ -ve.

bit size = 3

$(-3)_{10} = (?)_2$

$\downarrow$

MSB

$(3)_{10} = (011)_2 \xrightarrow{2^{nd}} (101)_2$

$\downarrow$ 1's comp

$\sim(3)_{10} = 100$

3 bit

$(-4)_{10}$

$\downarrow$

MSB

$(4)_{10} = (100)_2$

$\downarrow$

2'd comp X

$(101)_2$ its negative representation of 3 i.e -3

Range $= [-2^{n-1}, \; 2^{n-1}+1]$

for 4 bit Range $\to [-8, 7]$

1000

Qs Check the given number is odd or even using binary Operation

num1 = 22 MSB → $\underline{1}011\underline{0}$ ← LSB

num2 = 31 11111

$(13)_{10} = (1101)_2$

$(10)_{10} = (1010)_2$

for even number LSB = 0

for odd number LSB = 1

64 bit

$(22)_{10} = (10110)_2$

$(1)_{10} = (00001)_2$

even

num & 1 = 0

$(16)_{10} = (\underbrace{000000 \ldots 0\ 10110}_{64\ \text{bit}})_2$

& $(1)_{10} = (0000 \ldots\ldots 0001)_2$

$(000 \ldots\ldots 000)_2 = (0)_{10}$

64 bit New

32 bit

C int n = 5

odd

num & 1 = 1

8 bit

$(31)_{10} = (000\ 11111)_2$

& $(1)_{10} = (00000001)_2$

$(00000001)_2 = (1)_{10}$

Left Shift

$128 + 64 + 14 = (206)_{10}$

11001110 << 2

increase

0 0 ... 0	1	1	0	0	1	1	1	0

0 ... 0	1	1	0	0	1	1	1	0	0	0

$$\boxed{num \ll shift \Rightarrow num * 2^{shift}}$$

$(6)_{10} \rightarrow (00110)_2$

$(00011000)_2 = (24)_{10}$

16 8

Right shift

$(25)_{10} >> 2$

$16+8+1$

$(25)_{10} = (00011001)_2 >> 2$

$(25_{10}) >> 2 = 6$

$25/4 = \lfloor 6\ldots \rfloor$

0	0	0	1	1	0	0	1
0	0	0	0	0	1	1	0

$= 6$

4 2

$$\boxed{num >> shift = \left\lfloor num / 2^{shift} \right\rfloor}$$

0 → Ans the num (new num)

00010101

3rd bit

3rd bit = 1

Q any Number will be given, finds

(i) Get ith bit → 0, 1

(ii) Set ith bit → new number [ith bit = 1]

(iii) clear ith bit. [ith bit = 0]
↳ new number

3rd bit

00010101
& $\boxed{00000100}$?

00000100 >> 2

4th bit

00010101
& $\boxed{00001000}$?

$(00000000)_2 = (0)_{10}$ → bit = 0

bit mask → Magical Num = $1 << (i-1)$

ith bit = ?

Given Number & Magical Number = Result $>> (i-1)$

0 → bit 0

tve → bit 1

0 1

day-8

integer

num = 10100100 ← set(3)

base 10

00001000 ← bit mask (magical num)

(or) 10101100

$1 << i$

$(00000001)_2 << 3$

$= (00001000)_2$

$$\boxed{\text{Set bit} = num \mid (1 << i)}$$

Clear ith bit

10011100 ← 0, 2nd bit ← 3rd position

and (&) 11111011 ← bit mask

bitmask $\sim(1 << i)$

$\sim[(00000001)_2 << 2]$

$\sim(00000100) = 11111011$

10010010

11101111

10000010

Clear last i bit

num = 10011011

ans → 10010000

i = 4

$$\boxed{ans = num \;\&\; [(\sim 0) << i]}$$

num = 10011011

and(&) 11110000 ← bit mask.

$\therefore$ $\boxed{bitmask = (\sim 0) << i}$

$(\sim 0) = -1$

$= 11111111$

$-1 << 4 = 11110000$

Clear Range of bit (i, j)

num = 1 0 $\boxed{1\,1\,0\,0}$ 1 1

0 0 0 0

j → 5, i ← 2

bit mask = 11 0 0 0 0 11

= 1 1 0 0 0 0 0 0 (a)

0 0 0 0 0 0 1 1 (b) = 3

or. 11 0 0 0 0 11 ← bit mask

$a = (\sim 0) << (j+1)$

$(\sim 0) = 1111\ 1111$

$(\sim 0) << (5+1) = 1100\ 0000$

1	1	1	1	1	1	1	1
1	1	1	0	0	0	0	0

b = 0 0 0 0 0 1 1 1 ← 7

0 0 0 0 1 1 1 1 ← 15

$2^3\ 2^2\ 2^1\ 2^0$

$15 = 2^0 + 2^1 + 2^2 + 2^3$

$= 2^4 - 1$

$\boxed{b = 2^i - 1}$

$\boxed{b = 1 << i - 1}$

1 << 2 − 1

0 0 0 0 0 1 0 0 = 4 − 1 = 3

$$\boxed{\text{bitmask} = [(\sim 0) << j+1] \mid [(1 << i) - 1]}$$

Ans = num & bitmask.